

Who Counts? Implications of Competing Definitions of Career and Technical Education Students for Research Findings

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Abstract

Objective. Researchers use a range of operational definitions to identify career and technical education (CTE) students in administrative data, yet little is known about how these choices influence empirical findings. This study examines how differing identification approaches shape the observed prevalence, demographic characteristics, and academic and labor market outcomes of CTE students at community colleges. **Methods.** Using administrative records from the Ohio community college system for nearly 350,000 first-time community college students, we implemented multiple identification approaches based on stated preferences (e.g., declared major, career-focused educational intentions) and revealed preference definitions (e.g., number of CTE credits, share of credits in CTE fields). **Results.** Identification approaches markedly altered who was classified as a CTE student. Stated preference approaches identified larger groups of students who were younger, more likely to be female, and showed conventional positive earnings premia for credential completion. Revealed preference approaches yielded smaller, older cohorts with higher concentrations of CTE coursework. Within this group, the threshold rule mattered: under proportional credit thresholds, non-completing students exhibited higher observed earnings than completers five years after entry, whereas fixed credit thresholds produced more conventional positive earnings premiums for credential completion. Consequently, estimated CTE credential earnings premia varied across the approaches. **Contributions.** The findings demonstrate that seemingly routine operationalization decisions can substantially influence descriptive and earnings-based conclusions about CTE participation. We offer practical guidance for researchers and policymakers using administrative data and contribute to broader methodological debates about how measurement choices shape findings in postsecondary education research.

Keywords: community college; career and technical education; administrative data; operational definitions; credential completion; labor market outcomes.

Who Counts? Implications of Competing Definitions of Career and Technical Education

Students for Research Findings

Students in career and technical education (CTE) programs—such as health, business, information technology, protective services, and manufacturing (NCES, n.d.-a, n.d.-b)—account for a considerable share of enrollments at sub-baccalaureate institutions, particularly community colleges (NCES, 2016). Nearly two-thirds of first-time postsecondary students at community college are CTE majors (NCES, 2016). Policymakers increasingly view CTE programs as important mechanisms for addressing local labor-market needs (Atwell et al., 2022; Autor, 2014) and promoting economic mobility (Carruthers & Jepsen, 2021; Soliz, 2023), as reflected in the recent extension of federal Pell Grant eligibility to short-term workforce programs (U.S. Department of Education, 2026).

A growing body of literature reveals both the promise and challenges facing CTE in community colleges. CTE students tend to be older and more likely to work full-time than their peers (Hirschy et al., 2011). Many studies demonstrate positive labor market returns to CTE credentials, particularly associate degrees and longer certificates (e.g., Carruthers & Jepsen, 2021; Soliz, 2023; Stevens et al., 2019). Yet, stopping out remains common (Crosta, 2014; Jenkins & Cho, 2012), and students who complete only a few CTE courses typically experience smaller economic gains than credential completers (e.g., Bahr, 2019; Schudde & Shea, 2022).

However, identifying who qualifies as a CTE student is less straightforward than it may appear. Some researchers rely on stated preferences, such as declared major or self-reported academic intentions collected near the time of first enrollment (e.g., Bahr et al., 2025; Sublett & Plasman, 2024). Others rely on revealed preferences, such as the number of credits they attempt or complete in CTE fields (e.g., Bahr et al., 2023; Sublett & Plasman, 2025; Stevens et al., 2019).

or whether students complete a credential in a CTE field (e.g., Sublett & Plasman, 2025; Stevens et al., 2019). These differences reflect variation in available data and research objectives.

Different approaches could produce markedly different samples of students, meaning that findings about CTE students' characteristics, behaviors, and outcomes may not be comparable across studies. This confounds efforts to build empirically grounded policies and practices. When preparing data for analysis, researchers exercise considerable latitude in operationalizing concepts like "CTE student," and these choices may be consequential in ways not readily apparent (Huntington-Klein et al., 2021). Understanding how identification decisions shape findings is essential for methodological rigor and evidence-based policymaking.

In this study, we examine how differing approaches to identifying CTE students shape their observed prevalence, demographic characteristics, and academic and labor market outcomes. Using administrative data from Ohio that include nearly 350,000 first-time community college students, we implement a range of identification approaches based on information commonly available in administrative data. The approaches include stated preferences (declared major, academic intention) and revealed preferences (number of course credits in CTE fields, share of course credits in CTE fields). Beyond descriptive comparisons, we estimate regression models of post-college earnings for each analytic sample to assess whether operationalization choices alter conclusions about the earnings associated with credential completion.

Our findings reveal that different approaches capture distinct groups of CTE students that vary markedly in size, composition, and subsequent economic returns. Differences between stated preference and revealed preference approaches tend to be especially pronounced. Stated preference approaches yield the largest groups and identify students who are younger, more likely to be female, and have lower pre-college earnings. These students account for most CTE credential earners and

exhibit the conventional positive earnings premium for credential completion. Revealed preference approaches tend to isolate smaller, older cohorts who are disproportionately male, have higher pre-college earnings, and exhibit intensive CTE course-taking patterns. Under the revealed preferences framework, both the threshold types and measurement timeframe are consequential. While proportional thresholds identify early CTE concentrators whose non-completing students exhibit higher observed earnings than completers, fixed credit thresholds capture a broader group with more conventional post-college earnings patterns.

Our results show that a seemingly straightforward decision—how to operationalize CTE students—substantially influences both descriptive and earnings-based conclusions. This study offers practical guidance for researchers, administrators, and policymakers using administrative data to study CTE participation, contributes to a broader methodological literature on how researchers’ analytical decisions shape empirical findings (Huntington-Klein et al., 2021), and speaks to broader challenges of defining student participation in postsecondary research.

Background

Postsecondary CTE in Community Colleges

CTE, sometimes referred to as occupational or vocational education, is a longstanding fixture of the U.S. educational landscape (Bahr & Gross, 2023; Carruthers & Jepsen, 2021). For purposes of federal CTE reporting and this study, postsecondary CTE is defined as sub-baccalaureate programs of study that provide the requisite knowledge and skills for specific jobs or occupational fields (NCES, n.d.-a; Perkins V, 2018), such as business, consumer services, health sciences, manufacturing, and protective services (NCES, n.d.-a). Postsecondary CTE is delivered by a variety of institutions (Perkins V, 2018), but this analysis focuses on community college CTE.

Official data sources offer varying portraits of CTE's prevalence in community colleges. When focusing on students, *BPS 2012* reports that two-thirds (64%) of students who began postsecondary education at public community colleges are declared or intended CTE majors, including 62% of students pursuing associate degrees and 87% of students pursuing certificates (NCES, 2016). Alternatively, when focusing on credentials conferred, over half of credentials issued by community colleges (56%) are workforce or career-technical credentials, including 35% of associate degrees and 81% of certificates (Jenkins et al., 2025). This variation stems in part from different measurement approaches, either of which can be valid depending on research objectives.

Researchers can use two types of information to identify CTE students: stated preferences and revealed preferences (e.g., Zeidenberg et al., 2015). Stated preferences encompass what students say they are seeking to accomplish, such as declared major or reported academic intention. Revealed preferences encompass the goals that can be discerned from students' behaviors, including course enrollments or credentials completed, as reflected in the transcripts and award records maintained by institutions.

Stated Preferences, or “Trust What They Say”

The most straightforward approach is documenting whether students' declared majors fall into fields of study categorized as CTE (Sublett & Plasman, 2024; Sublett & Tovar, 2021). However, major information may be unavailable, incomplete, or a trailing indicator of students' true goals (e.g., Bahr et al., 2025; Liu et al., 2021; Zeidenberg et al., 2015). An alternative is self-reported academic intentions, encompassing students' goals, plans, or aspirations regarding their educational pathways, such as credential objectives and immediate workforce goals (Bailey et al., 2006; DesJardins et al., 2019). In community colleges, students seeking immediate workforce preparation are most often served by CTE programs, making stated intentions a useful proxy for

field of study (Bahr et al., 2025). However, intentions may change over time, may not be collected by colleges consistently or at all (Bailey et al., 2006; Brand et al., 2014), and may not reflect students' true goals (Hirschy et al., 2011). Moreover, unlike declared majors, which can reference established classification standard (e.g., CIP codes), academic intentions do not lend themselves to consistent measurement across settings, limiting comparability across studies.

Revealed Preferences, or “Trust What They Do”

Students' choices or behavior also can communicate their educational goals. Behavioral measures reflect time, effort, and financial investments in achieving particular goals and are at least as indicative of those goals as self-reported measures.

Identifying CTE students by credential field (e.g., Sublett & Plasman, 2025) is limited by the comparatively low completion rates at community colleges (Zeidenberg et al., 2015), which leave the CTE status of non-completing students undefined. This measure also trails enrollment, limiting its utility for studying contextual and experiential factors in college.

CTE students also can be identified by their course-taking patterns (Dougherty et al., 2020). Some researchers apply fixed credit thresholds, defining CTE students as those who attempt a specified number of CTE credits, such as three, six, and nine credits (one to three courses) (e.g., Columbus, 2025; Jenkins & Cho, 2012). Research on non-completing students, a significant proportion of community college students, consistently use comparable credit thresholds to identify comparison groups (Bahr, 2019; Foote & Stange, 2022; Stevens et al., 2019). Fixed thresholds measure an absolute level of CTE course-taking, therefore capturing students with sustained participation in CTE, regardless of whether they also pursue coursework in other fields.

Others apply proportional credit thresholds, defining CTE status by the share of all credits in CTE fields (e.g., Bahr et al., 2023). Proportional thresholds identify students who concentrate

their educational effort in CTE fields regardless of their overall course load. This approach accommodates the diverse enrollment patterns common in community colleges, where many students enroll part-time or attend college episodically (Crosta, 2014) and may be more effective at identifying students in short CTE certificate programs.

These approaches capture students' CTE course-taking in different ways. The implications of identifying CTE students through fixed versus proportional credit thresholds are largely untested. Both are subject to complexities related to the window of time in which to determine CTE status. Shorter windows, such as the first year of enrollment, are less susceptible to attrition bias and more valuable for assessing early academic momentum, which is predictive of short-run persistence and long-run outcomes (Jenkins & Bailey, 2017; Wang, 2017). Nevertheless, CTE intentions may poorly reflect CTE intentions if institutions prescribe course sequences; for example, students may complete development education before taking major courses (CCRC, 2021). Longer windows, such as four years after entry, allow behaviors to reveal intentions more reliably but are vulnerable to attrition bias resulting from students leaving college during the observation window (Crosta, 2014; Jenkins & Bailey, 2017).

This Study

To investigate whether and how CTE identification approaches produce different analytic conclusions, we compared eight operationalizations of CTE students drawn from two broad categories: stated preferences (students' declared major and self-reported academic intentions) and revealed preferences (students' course-taking behavior, measured via proportional and fixed credit thresholds). Each operationalization was applied to the same underlying sample of community college entrants, allowing us to examine how the choice of definition affects the prevalence,

characteristics, and outcomes of the resulting CTE student samples. We describe the data, identification approaches, and key measures below.

Data

Source. We use administrative data from the Ohio Longitudinal Data Archive (OLDA), which includes data from all public postsecondary institutions in Ohio as well as matched records from the state's Unemployment Insurance (UI) database and the National Student Clearinghouse (NSC).¹ The OLDA dataset includes information on student demographic characteristics and enrollment, transcripts, credentials awarded, and quarterly earnings records for students who attended any of the 23 public community colleges in Ohio. Importantly, it includes all courses attempted by students, as well as their declared majors and self-reported academic intentions in their first term of enrollment. The UI data permit us to investigate students' earnings and employment histories both prior to enrolling and after exiting community college. The NSC data enable us to track student transfers to other postsecondary institutions, although that is not an explicit focus of this analysis.

Sample. We constructed a sample of 342,660 students whose first observed term of college enrollment was at an Ohio community college between July 1, 2006, and June 30, 2011.² We limited our sample to working-age students 18 to 60 years old at entry, who were not dual-enrolled in high school and community college, and who had complete information on gender and age at entry. We excluded students who lacked a workforce identifier required to link records to UI data. Selected characteristics of this sample are provided in Table 1 (Column 1).

[Insert Table 1 about here]

Identification Approaches

We investigated multiple approaches to identifying and distinguishing CTE students from the rest of the community college student population. These approaches can be broadly categorized as being based on stated preferences or revealed preferences. Stated preference approaches draw on students' declared major field of study or their self-reported academic intentions. Revealed preference approaches draw on students' course-taking, such as the number of course credits that students attempted in CTE fields or the share of all course credits attempted by students that were in CTE fields. We used the NCES Postsecondary Taxonomy to categorize fields as being CTE or non-CTE (NCES, n.d.-b), which is widely used in research on postsecondary CTE (e.g., Sublett & Plasman, 2025) and in federal statistics (e.g., NCES, 2016). The taxonomy identifies 26 two-digit Classification of Instructional Programs (CIP) codes as CTE.³

Stated Preferences. We used two stated preference approaches to identify CTE students, each based on distinct measures collected in students' first term of enrollment—declared major field of study and self-reported academic intention. We recoded major field of study from its original two-digit CIP classification into a binary variable distinguishing CTE from non-CTE programs. We similarly collapsed the ten-category academic intention variable, originally ranging from earning a bachelor's degree to acquiring knowledge for personal interest, into a binary measure: terminal career-focused intentions versus all other intentions. The four terminal career-focused intentions were: obtain a postsecondary certificate, obtain an associate degree for the job market, train for a new career by taking only selected courses, and upgrade skills for a current job by taking only selected courses.

Revealed Preferences. We constructed two sets of revealed preference indicators using information about the courses in which students enrolled, providing consistency with established

research practices. The first set of three indicators focused on whether students attempted a specified minimum number of course credits in CTE fields within their first year in community college. We examined three thresholds—three, six, and nine CTE course credits, respectively, which approximate enrolling in one, two, and three three-credit CTE courses. The second set of three indicators focused on whether the percentage of all course credits attempted by students in their first year that were in CTE fields met or exceeded a specified threshold. Again, we examined three thresholds—one-third (33%), half (50%), and two-thirds (66%). Students whose share of CTE credits met or exceeded a given threshold were identified as CTE students. Developmental education credits are excluded from CTE credits but retained in total credits. Consequently, coursework in developmental education lowers a student’s CTE credit share rather than removing the student from the sample, preserving realistic course-taking patterns and avoiding attrition across measures.

Key Measures

Background characteristics. We examined a set of background characteristics including gender, age at entry, race/ethnicity, and pre-college earnings. Gender is measured as a binary variable. Age is measured at initial enrollment. The categories of race/ethnicity include White, Black/African American, Hispanic/Latino, Asian American and Native Hawaiian/Pacific Islander (AANHPI), Other, and Unknown/Missing.⁴ Our measures of pre-college employment and earnings are based on matched UI wage data from the eight fiscal quarters (two years) prior to a student’s first enrollment in an Ohio community college. We calculated two measures of students’ average quarterly earnings—a *conditional* measure excluding quarters with real earnings (2020Q1 dollars) below \$100, and an *unconditional* measure including all quarters regardless of earnings amount.

Outcomes. We measured CTE prevalence as the share of the full sample identified as CTE students under each approach. We examine how different definitions of CTE students influence findings on several educational and labor market outcomes, including credential completion, credit accumulation, and post-college employment and earnings.

We define credential completion as the attainment of any community college credential—postsecondary certificate or associate degree—from an Ohio community college within four years of initial enrollment, which represents 200% of the normal time to complete an associate degree (NCES, n.d.-c). Many community college students attend part-time or episodically, hence the extended observation period for the outcome of completing a postsecondary credential (Crosta, 2014; Hirschy et al., 2011; Jenkins & Cho, 2012). We also distinguish completing a credential in any field from completing a credential in a CTE field specifically, as determined by the NCES Postsecondary Taxonomy. As a validation, we calculated what proportion of all CTE completers in the full sample each definitional approach captured to reveal the comprehensiveness of each approach in identifying CTE completers. Students who transferred to any four-year institution prior to credential completion are excluded from all measures of credential attainment. We measured students' credit accumulation at four years after college entry, including total community college credits attempted in all fields and the share of credits attempted in CTE fields, which capture overall academic progress and CTE intensity, respectively.

In regard to post-college employment and earnings, we calculated each student's average conditional and average unconditional earnings in the fifth year, applying the definitions of conditional and unconditional described above. We also calculated descriptive earnings premia by differencing average earnings of completers and non-completing students, recognizing the

differential returns to completion versus non-completion observed in prior research (Bahr, 2019; Carruthers & Jepsen, 2021; Zeidenberg et al., 2015).

Regression Estimates. To examine whether CTE operationalization decisions affect substantive research conclusions, we estimated an earnings premium regression for each analytic sample. Specifically, we regressed average unconditional quarterly earnings in Year 5 on a three-category credential completion variable distinguishing students who completed a CTE credential, completed a non-CTE credential, or did not complete any credential within four years of community college entry (the reference group). Additional covariates included an indicator for enrollment in any postsecondary institution during Year 5, whether transferred to four-year institutions, pre-college average unconditional quarterly earnings, age at entry, age at entry squared, gender, and race/ethnicity. All models were estimated using ordinary least squares with robust standard errors. These estimates are descriptive, adjusted associations and should not be interpreted as causal effects of credential completion. The two coefficients of primary interest—the estimated earnings premia associated with CTE and non-CTE credential completion relative to non-completion—were extracted from each model and compiled into a summary table to facilitate cross-sample comparison.

Results

We organize our analysis into three parts. We first compare CTE students identified through the two stated preference approaches—declared major versus career-focused academic intentions in the first term. We then compare CTE students identified through the six revealed preference approaches, contrasting the proportional threshold approaches (33%, 50%, 66% of credits in CTE fields) with the fixed threshold approaches (3, 6, 9 CTE credits), measured in the first year. Third, we consider the broader similarities and differences in the CTE students

identified through stated versus revealed preference approaches. Following descriptive statistics (Table 1, Panel A), we present regression analyses demonstrating how identification method choice alters estimated earnings premia (Table 1, Panel B). Across all eight samples, the estimated CTE credential premium is positive and significant ($p < .001$), ranging from \$528 to \$1,552 per quarter (2020Q1 dollars), while the estimated non-CTE credential premium is generally negative or not statistically significant.

Stated Preferences: Declared Major versus Academic Intention

Overall, the two measures of students' stated preferences—declared major and career-focused academic intentions—produced similar samples in terms of demographics and completion outcomes, though the declared major group was slightly younger, had lower pre-college earnings, included more CTE students, and captured more eventual CTE completers.

As shown in Table 1 (Column 2-3), the declared major approach identified a larger share of enrollees as CTE students (66%) than the academic intentions approach (59%) among our full sample of community college entrants ($N = 342,660$). The prevalence rates in both samples approximate a reference point of 64% from *BPS 2012* (NCES, 2016). The samples differed only modestly in demographics and completion outcomes. The declared major sample was younger (49% aged 25 or older versus 54%) with lower pre-college quarterly earnings (\$5,134 versus \$5,606).

Completion rates were nearly identical (13% each), with approximately 90% of completers in both samples attaining CTE credentials. The declared major sample captured 85% of all eventual CTE completers in the full sample, compared to 73% for the academic intentions sample—a difference partially attributable to the declared major sample's higher prevalence. Credit accumulation patterns were similar: with comparable total credits (45 vs 43) and CTE credit concentration (51% vs 53%) across samples.

Earnings outcomes revealed larger differences between samples both descriptively and in regression analyses). Descriptively, the declared major sample showed a larger earnings premium for completion (\$1,339 versus \$1,094). However, the declared major sample had a smaller estimated earnings premium for CTE credentials (\$1,462 vs. \$1,552, $p < .001$), despite showing little penalty for non-CTE credentials (-\$19, $p > .10$, versus -\$183, $p < .05$), when controlling for background characteristics and enrollment status at the time of measurement (Table 1, Panel B).

Finally, the samples overlapped largely but not completely: 68% of declared CTE majors reported career-focused intentions, while 76% of those with career-focused intentions declared a CTE major (Table 2, Column 1-2). Given that the declared major criterion produced a more prevalent sample that captured a higher proportion of eventual CTE completers, we selected it as the primary stated preference measure for subsequent comparisons.

[Insert Table 2 about here]

Revealed Preferences: Fixed versus Proportional Credit Thresholds

Proportional and fixed credit thresholds produce samples that vary greatly in the prevalence of CTE students (Table 1, Column 4-9), although all fall below the BPS benchmark of 64% (NCES, 2016). Proportional thresholds identify CTE students ranging from 53% to 33% of all enrollees (Column 4-6). Fixed thresholds exhibit more dramatic variation in prevalence, ranging from 62% to 28% (Column 7-9).

Proportional and fixed credit thresholds identify markedly different groups of CTE students. Proportional thresholds consistently identified older students (ages 31-33 versus 29 for fixed thresholds) with higher pre-college earnings (\$6,342-\$7,581 versus \$5,172-\$5,274). For both threshold types, female representation declined as criteria became more restrictive (proportional: 43% to 37%; fixed: 49% to 44%). Racial and ethnic composition remained stable across all criteria.

Students identified through proportional credit thresholds had noticeably lower completion rates than those identified through fixed thresholds (11-13% versus 15-24%). All first-year identification approaches effectively identified students who eventually completed CTE credentials. At comparable prevalence levels, proportional thresholds captured smaller shares of eventual CTE completers than fixed thresholds (58% for the 50% sample versus 80% for the 3-credit sample), but more restrictive thresholds captured notably fewer eventual CTE completers in the full sample (e.g., 38% for the 66% samples). Despite attempting fewer total credits (31-40 versus 50-63 credits), students in proportional threshold samples exhibited higher CTE credit concentration (74-88% versus 62-69%).

The largest difference between the threshold types involves post-college earnings. Descriptively, non-completing students identified through proportional thresholds consistently earned more than those identified through fixed thresholds (\$8,593-\$9,840 versus \$7,500-\$7,558). Proportional threshold samples demonstrated minimal or negative descriptive earnings premiums for completers. In particular, the 50% and 66% samples had negative earnings premiums (-\$146 and -\$501), meaning non-completing students outearned completers, on average, five years after entry. These descriptive earnings gaps diminished in regression analyses (Table 1, Panel B), although the proportional samples still had much lower estimated CTE credential premium than the fixed samples (\$528-\$862 versus \$1,146-\$1,253, $p < .001$). These patterns indicate that proportional thresholds disproportionately identified non-completing students, including many who out-earned completers.

The overlap among revealed preference samples illustrates the trade-offs underlying proportional versus fixed thresholds (Table 2, Columns 3-8). Across the proportional threshold samples (Columns 3-5), the percentages of students attempting at least 3, 6, or 9 CTE credits

remain remarkably stable even as the thresholds become more restrictive. Conversely, most students in the fixed threshold samples also exhibited high proportions of CTE credits (Column 6-8), but the more restrictive fixed thresholds capture considerably smaller samples, limiting their practical utility for first-year measurement.

Stated Preferences versus Revealed Preferences

When comparing the least restrictive samples across stated versus revealed preferences (declared major versus 3-credit; see Table 1, Column 2 and 7), the declared major sample was slightly larger (66% versus 62%) and younger (28 versus 29), with lower CTE credit concentration (51% versus 62%). The samples were similar in their completion rates (13% each) and coverage of eventual CTE completers (85% versus 90%). Differences between stated and revealed preference samples were more pronounced at more restrictive thresholds.

The samples showed more notable differences in post-college earnings. Non-completing students in the 3-credit sample outearned those in the declared major sample (\$7,558 versus \$7,304). The descriptive earnings premium for completion was also smaller in the 3-credit sample (\$900) compared to that of the declared major sample (\$1,339) —a comparison aligning with the estimated earnings premia of both CTE credentials (\$1,253 versus \$1,462, $p < .001$) and non-CTE credentials (-\$183, $p < .01$, versus -\$19, $p > .1$; see Table 1, Panel B).

The overlap between the two samples was similar to that of the two stated preference samples: 70% of declared major students attempted 3 or more credits in CTE fields in the first year (Table 2, Column 1), while 74% of the 3-credit sample declared CTE majors (Table 2, Column 6).

Discussion

As research on postsecondary CTE has expanded, scholars have used diverse approaches to identify CTE students, ranging from stated preference measures (e.g., majors and academic

intentions) to revealed preferences based on course-taking and credential attainment. Yet little evidence has examined whether these choices meaningfully affect empirical results.

In our analysis, identification approach substantially shaped who was classified as a CTE student. Stated preference measures identified a broad population that was younger, more likely to be female, and captured most eventual CTE completers. In contrast, revealed preference measures identified smaller groups characterized by greater labor market attachment, stronger CTE concentration, and different earnings trajectories. Non-completing students identified through revealed preference measures were much more intensively engaged in CTE coursework. These differences were most pronounced for proportional threshold rules, which appeared to isolate early CTE specialists and non-completing students with higher observed earnings than completers five years later, particularly at more restrictive thresholds.

One challenge in conducting research on CTE students is that researchers often must identify these students using only limited information, typically information collected in the first term of enrollment. In such cases, the appropriate criterion depends on the research purpose. Our results demonstrate that the declared major approach offers several advantages: its estimated prevalence closely aligns with national benchmarks (64%), it captures the largest share of eventual CTE completers (85%), and it produces earnings patterns consistent with prior literature (Carruthers & Jepsen, 2021). Practically, it is also widely available and straightforward to implement in administrative datasets. These advantages, however, come with documented trade-offs (e.g., Zeidenberg et al., 2015). Declared majors appear to overidentify CTE students: almost one-third of declared CTE majors in our sample attempted fewer than three CTE credits during their first year. Researchers may therefore benefit from combining major declarations with minimum course-taking requirements.

When using first-year course-taking to identify CTE students, the choice between threshold types involves a tradeoff. Proportional thresholds tend to identify highly specialized CTE students, potentially excluding students whose broader course-taking reflects prerequisites, general education requirements, or uncertainty about program entry. Fixed thresholds, by contrast, set a lower bar for CTE engagement and capture students with more varied course-taking profiles.

A noteworthy pattern is that non-completing students in proportional threshold samples generally outearn completing students five years after entry. One possible explanation is that these samples disproportionately capture skills builders, a distinctive group of non-completing students, generally representing 11-14% of community college enrollees across states, who enroll briefly in targeted CTE coursework and subsequently realize meaningful earnings gains after exiting college (Bahr, 2019; Bahr et al., 2023). The strong CTE concentration required by proportional thresholds closely aligns with this pattern of participation. Because research has documented returns to credits in specific CTE fields for non-completing students (e.g., Bahr, 2019; Schudde & Shea, 2022), non-completing students in the proportional threshold samples may disproportionately enroll in high-return CTE fields. Additionally, these students might also disproportionately attain third-party credentials such as certifications or licenses (Ewert & Kominski, 2014), which are rarely tracked in postsecondary administrative data (Cardenas-Navia, 2021; Van Noy, 2020) and can likewise result in earnings gains (Baird et al., 2021; Xu et al., 2024).

Our results have implications for policy and practice. Evaluations of the economic benefits of CTE programs for individuals and localities influence the policies that support and fund CTE provision, and accountability dashboards often rely on metrics of CTE participation and success (e.g., Advance CTE and the Association for Career and Technical Education, 2020; DeLoach et al., 2023; Joseph & Canney, 2019). Our findings suggest that commonly used indicators of CTE

participation and success may not be robust to alternative identification strategies. Agencies and system offices may therefore benefit from evaluating existing metrics under multiple definitions or adopting composite measures that incorporate both stated intent and demonstrated participation. Relatedly, the observed misalignment between intent and course-taking patterns also points to opportunities for institutional interventions. Cross-checking academic intentions with registration behavior could help identify students experiencing barriers to program entry and support early academic momentum (Jenkins & Cho, 2012). More generally, declared major may be most useful for broad early identification and advising, fixed credit thresholds for demonstrated exposure to CTE coursework, proportional thresholds for identifying CTE concentration or skills-builder patterns, and credential field for completion-focused accountability.

Our analysis is not without limitations. It is confined to community college students in Ohio who first enrolled many years ago. The UI wage data also do not capture self-employment, federal employment, out-of-state employment, or other employment outside the covered wage-record system. Replications across multiple states using more recent data would immediately advance this line of work. We also do not explore the full array of stated preference or revealed preference measures that one could employ to identify CTE students. One could use advanced predictive methods to identify CTE students based on many more dimensions than we considered in this analysis (e.g., Bird et al., 2024; Zeidenberg et al., 2015). We also do not examine an exhaustive set of background characteristics and outcomes that one could feasibly construct using available administrative data, and the variation in estimated earnings premia across identification approaches warrants further investigation using causal designs. An important extension would examine whether alternative definitions produce similar patterns in designs that address selection

bias more directly (Ross et al., 2020), including fixed-effects models of returns to CTE participation and credential attainment (Belfield & Bailey, 2017).

These insights extend beyond CTE. Researchers studying STEM, liberal arts, developmental education, and specific CTE fields (e.g., Bahr, 2012, 2013; Holzer & Xu, 2021; Jenkins & Cho, 2012; Jenkins & Weiss, 2011; Wang, 2016) likewise assign a field of study to individual students in their datasets and often demonstrate methodological variability in defining target populations. Similar variability has also been noted in secondary CTE research (Dougherty et al., 2020; Goldring & Kreisman, 2024).

More systematic evidence is needed on how researchers identify CTE students and how those decisions shape findings. Definitional inconsistencies can complicate efforts to connect evidence across research and policy contexts and inform policy decisions regarding accountability, funding, and student support. Moving forward, researchers should clearly document their identification criteria and assess the robustness of findings to alternative definitions. This would be a meaningful step towards reducing the role of hidden researcher decisions in empirical evaluations of CTE students (Huntington-Klein et al., 2021). Greater transparency about identification decisions will strengthen the evidence based on CTE effectiveness and support policies that serve the diverse population of students pursuing career-focused education at community colleges.

Notes

¹ The Ohio Longitudinal Data Archive is a project of the Ohio Education Research Center (oerc.osu.edu) and provides researchers with centralized access to administrative data. The OLDA is managed by The Ohio State University's CHRR (chrr.osu.edu) in collaboration with Ohio's state workforce and education agencies(ohioanalytics.gov), with those agencies providing oversight and funding. For information on OLDA sponsors, see <http://chrr.osu.edu/projects/ohio-longitudinal-data-archive>.

² We define first enrollment on a term basis: students with any prior postsecondary enrollment observed in the data are excluded, so the sample reflects students beginning postsecondary education at an Ohio community college.

³ The NCES Postsecondary Taxonomy encompasses 26 fields of study at the level of two-digit CIP codes: Agriculture (01), Natural resources (03), Business and marketing (52), Communication (09), Communications technologies (10), Computer and information sciences (11), Family and consumer sciences (19), Personal and culinary services (12), Parks, recreation, and fitness (31), Education (13), Architecture (04), Engineering (14), Engineering technologies (15), Military technologies (29), Science technologies (41), Health sciences (51), Protective services (43), Military services (28), Legal professions and studies (22), Library science (25), Public administration and social services (44), Theology and religious vocations (39), Construction (46), Repair (47), Manufacturing (48), and Transportation (49) (NCES, n.d.-b).

⁴ Due to the small numbers of students in these categories, we aggregated the percentages of American Indians and Alaska Natives (AIAN) and Multiracial into the “Other” category. While nonresident status is treated as a valid race/ethnicity category, it provides no information about race/ethnicity of nonresidents; therefore, we coded Nonresidents into the “Unknown/Missing” category.

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Table 1. Prevalence, Characteristics, and Outcomes of Identified CTE Samples.

	Full Sample	Stated Preference		Revealed Preference					
		Declared Major	Academic Intention	Proportional Credit Thresholds			Fixed Credit Thresholds		
				33% of Credits	50% of Credits	66% of Credits	Three Credits	Six Credits	Nine Credits
N	342,660	225,921	201,962	182,567	149,240	113,511	212,257	142,005	97,525
Panel A. Descriptive Statistics									
Prevalence									
Identified CTE Students (%)	-	66%	59%	53%	44%	33%	62%	41%	28%
Background Characteristics									
Age at Entry (mean)	28	28	29	31	32	33	29	29	29
Aged 25 or Older (%)	47%	49%	54%	57%	60%	64%	49%	51%	52%
Female (%)	52%	51%	52%	43%	40%	37%	49%	46%	44%
White (%)	70%	69%	72%	70%	70%	70%	71%	72%	73%
Black/African-American (%)	19%	20%	18%	18%	18%	18%	18%	17%	16%
Hispanic/Latino (%)	2%	2%	2%	2%	2%	2%	2%	2%	2%
AANHPI (%)	1%	1%	1%	1%	1%	1%	1%	1%	1%
Other Race/Ethnicity (%)	2%	2%	2%	2%	1%	1%	2%	2%	1%
Unknown/Missing Race/Ethnicity (%)	6%	6%	6%	7%	7%	8%	6%	6%	6%
Unconditional Pre-College Quarterly Earnings (2020Q1\$) (mean)	4,129	4,071	4,452	4,981	5,381	5,995	4,109	4,009	3,992
Pre-College Quarterly Earnings (2020Q1\$) (mean) ^a	5,288	5,134	5,606	6,342	6,837	7,581	5,274	5,172	5,190
n (Employed, Pre-College)	267,606	179,119	160,398	143,396	117,458	89,755	165,380	110,063	75,015
Completion Outcomes in Four Years									
Credential Completion (%)	11%	13%	13%	13%	13%	11%	15%	20%	24%
CTE Credential Completers (%)	9%	12%	12%	13%	12%	11%	14%	18%	23%
All CTE Credential Completers Captured in Sample (%) ^b	100%	85%	73%	73%	58%	38%	90%	80%	70%
Total Credits Attempted (mean)	42	45	43	40	37	31	50	56	63
Percentage of CTE Course Credits Attempted (mean) ^c	46%	51%	53%	74%	81%	88%	62%	67%	69%
Post-College Earnings in Year 5 after Entry									
Unconditional Quarterly Earnings (2020Q1\$) (mean)	5,445	5,390	5,572	6,088	6,395	6,877	5,496	5,523	5,549
Quarterly Earnings (2020Q1\$) (mean) ^d	7,670	7,495	7,810	8,615	9,087	9,781	7,709	7,741	7,799
n (Employed, Post-college)	243,251	162,468	144,089	129,016	105,027	79,806	151,317	101,312	69,382
Quarterly Earnings for Completers (2020Q1\$) (mean)	8,363	8,644	8,751	8,745	8,961	9,339	8,458	8,639	8,709
n (Completers Employed, Post-college)	30,091	23,091	20,219	18,821	14,635	9,420	25,307	21,493	18,093
Quarterly Earnings for Non-Completers (2020Q1\$) (mean)	7,573	7,304	7,656	8,593	9,107	9,840	7,558	7,500	7,478
n (Non-Completers Employed, Post-college)	213,160	139,377	123,870	110,195	90,392	70,386	126,010	79,819	51,289
Descriptive Earnings Premium of Credential Completion (2020Q1\$) (mean)	791	1,339	1,094	152	-146	-501	900	1,140	1,230
Panel B. Estimated Earnings Premia of Credential Completion^e									

	Full Sample	Stated Preference		Revealed Preference					
		Declared Major	Academic Intention	Proportional Credit Thresholds			Fixed Credit Thresholds		
				33% of Credits	50% of Credits	66% of Credits	Three Credits	Six Credits	Nine Credits
Estimated Earnings Premium of CTE Credential (2020Q1\$)	-	1,462.480* **	1,551.784* **	862.824***	701.935***	528.097***	1,252.684* **	1,173.673* **	1,146.500* **
	-	(35.773)	(38.928)	(39.614)	(45.029)	(56.216)	(35.188)	(38.479)	(42.873)
Estimated Earnings Premium of Non-CTE Credential (2020Q1\$)	-	-18.953	-182.746**	52.038	-47.916	154.405	-182.838***	-19.339	142.984
	-	(91.798)	(89.281)	(128.553)	(179.276)	(263.716)	(70.566)	(96.212)	(124.910)

Notes. Each column represents a sample of identified CTE students constructed from the full sample of first-time Ohio community college enrollees ($N = 342,660$). Stated preference measures are based on the first term of enrollment; revealed preference measures are based on course-taking over the first year. All numbers are rounded. ^a Pre- and post-college quarterly earnings are averages in real 2020Q1 dollars (CPI-U adjusted), conditional on employment. “Pre-College Quarterly Earnings” is measured over the eight quarters (two years) preceding college entry. “n (Employed, Pre-College)” is the number of students with observed earnings in that window. ^b “All CTE Credential Completers Captured in Sample” represents the share of all CTE credential completers in the full sample who fall within each identified CTE sample. ^c “Percentage of CTE Course Credits Attempted” is the average share of total credits attempted in CTE fields at four years after college entry. Developmental education credits are excluded from CTE credits but retained in total credits. ^d “Post-College Quarterly Earnings” is measured over quarters 17-20 following college entry, disaggregated by credential completion status. “Descriptive Earnings Premium” is the difference in post-college earnings between completers and non-completers. ^e “Estimated Earnings Premium” are from ordinary least squares regressions of average unconditional quarterly earnings in Year 5 (fiscal quarters 17-20, 2020Q1 dollars) on a three-category credential completion variable (no credential [reference], non-CTE credential, CTE credential), Year 5 enrollment, transfer to a four-year institution, pre-college earnings, age at entry, age at entry squared, gender, and race/ethnicity. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table 2. Overlap of CTE Samples Identified Based on First-Term Stated Preference or First-Year Revealed Preference.

	Declared Major	Academic Intention	Proportional Credit Thresholds			Fixed Credit Thresholds		
			33% of Credits	50% of Credits	66% of Credits	Three Credits	Six Credits	Nine Credits
N	226,501	202,377	182,567	149,240	113,511	212,257	142,005	97,525
Declared Major	100%	76%	75%	75%	73%	74%	79%	83%
Academic Intention	68%	100%	67%	69%	69%	65%	68%	70%
Proportional: 33% of Credits	61%	61%	100%	100%	100%	77%	88%	94%
Proportional: 50% of Credits	49%	51%	82%	100%	100%	62%	71%	78%
Proportional: 66% of Credits	36%	39%	62%	76%	100%	46%	52%	56%
Fixed: Three Credits	70%	68%	90%	89%	86%	100%	100%	100%
Fixed: Six Credits	50%	48%	68%	68%	65%	67%	100%	100%
Fixed: Nine Credits	36%	34%	50%	51%	48%	46%	69%	100%

Notes. Each sample of identified CTE students is constructed from the full sample of all first-time enrollees in an Ohio community college (N = 342,660). Columns present identified CTE student samples; rows present overlapping definitional criteria. Example: 68% of the declared major sample (Column 1) reported career-focused educational intentions (Row 3), but only 50% attempted at least 6 CTE credits in their first year (Row 7).

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Statements and Declarations

Ethical considerations

This study uses restricted administrative data from the Ohio Longitudinal Data Archive (OLDA). The research was determined not regulated by the University of Michigan Institutional Review Board. All analyses were conducted in compliance with the data use agreement.

Declaration of conflicting interest

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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Consent to participate

Not applicable. This study uses de-identified and restricted administrative records.

Consent to publication

Not applicable.

Data availability

The data used in this study are drawn from the Ohio Longitudinal Data Archive (OLDA) and cannot be shared publicly due to data use agreement restrictions. Researchers interested in accessing OLDA data should contact the Ohio Education Research Center.